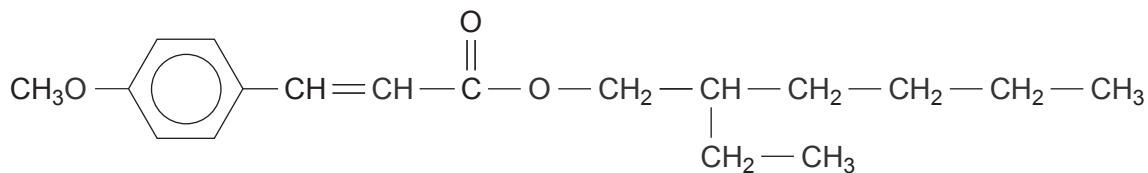
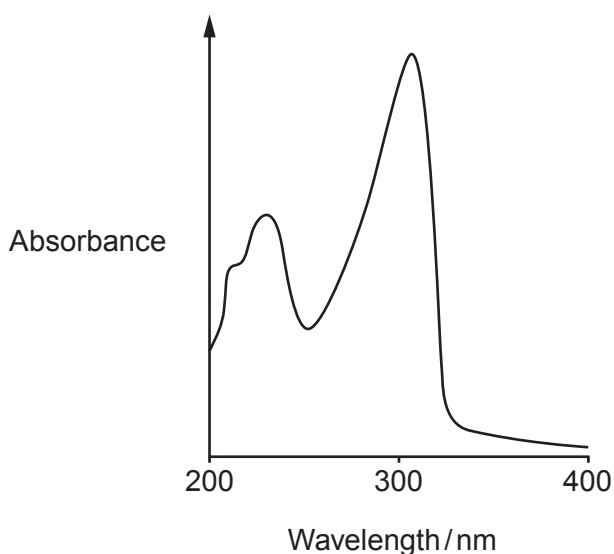


12. (a) People are increasingly concerned about the effect of the Sun's ultraviolet rays on the skin. A number of compounds have been developed for use as sunscreens. One of these compounds is compound **A**.

compound **A**

- (i) The ultraviolet absorption spectrum of compound **A** is shown below.



Ultraviolet radiation is divided into three regions.

Region	Wavelength/nm
UVA	320-400
UVB	290-320
UVC	200-290

Use this information to explain why it may be necessary to use another sunscreen compound, together with compound **A**, when producing a commercial sunscreen.

[1]

.....

.....



- (ii) Compound **A** is an ester that is made by reacting together 2-ethylhexanol and the appropriate acid, in the presence of a catalyst.

Give the molecular formula of the acid used in this reaction.

[1]

.....

- (iii) Compound **A** displays both forms of stereoisomerism.

I. State how structural isomerism differs from stereoisomerism.

[2]

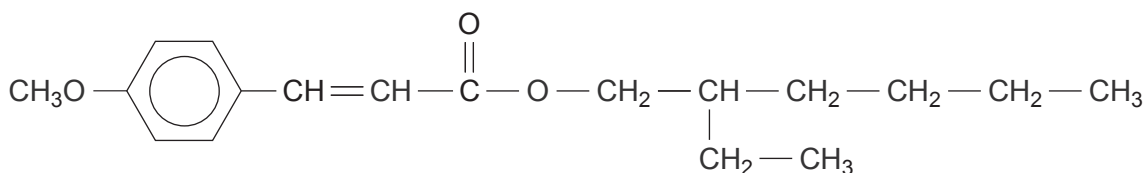
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II. Indicate the position of a chiral centre on the formula of compound **A** below.

[1]



- III. When compound **A** is made by the method described in part (ii) above, an equimolar mixture of both enantiomers is produced.

State how a solution of this mixture affects the plane of plane polarised light.

[1]

.....

- IV. Exposure of a mixture of both *E*- and *Z*- forms of compound **A** to UV radiation results in the *E*- form gradually changing to the *Z*- form.

Using the general formula of compound **A** as $\text{R}-\text{CH}=\text{CH}-\text{R}'$, draw and label these two forms of compound **A**.

[2]



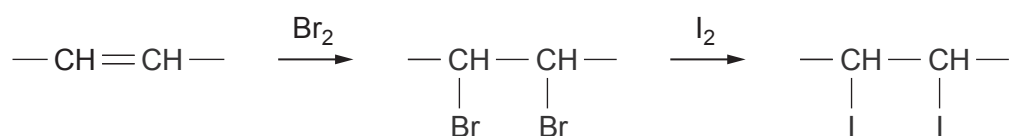
- (b) Palm oil contains around 40% of unsaturated oil and 60% of saturated oil. The amount of unsaturation in an oil can be measured indirectly in a reaction with iodine.

- (i) In the first stage of this method bromine adds across the $C=C$ double bonds present.

State the type of mechanism occurring when bromine adds across these double bonds. [1]

.....

- (ii) The overall reaction with iodine can be represented by the following equation.



In palm oil, most of the unsaturated oils are present as glyceryl trioleate (M_r 885). This compound contains three $C=C$ double bonds per molecule.

An 8.41 g sample of palm oil reacted indirectly with 0.0128 mol of iodine (I_2).

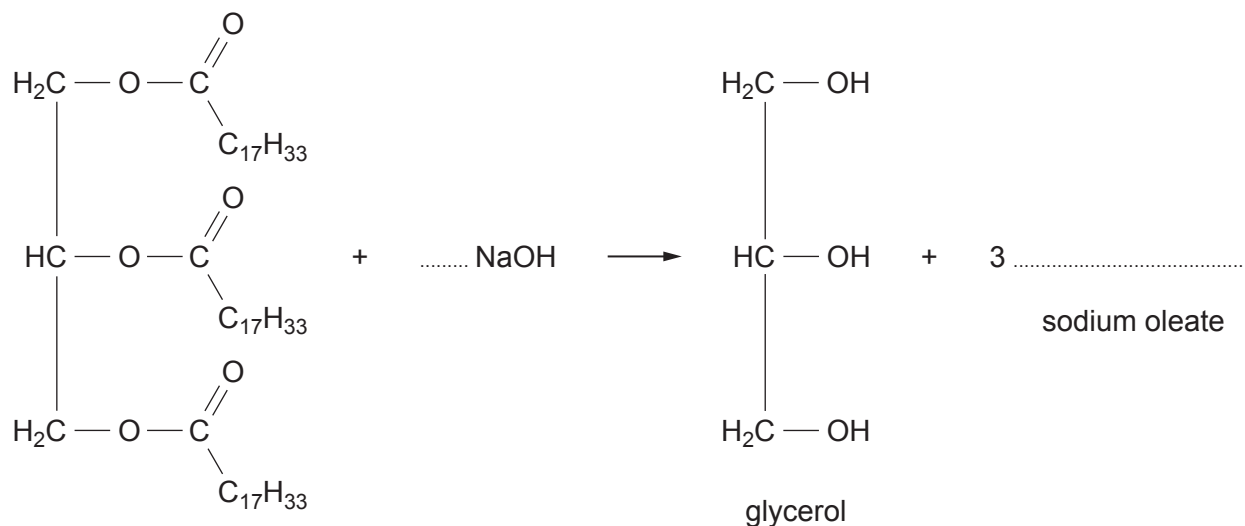
Calculate the percentage of unsaturated oil (as glyceryl trioleate) present in the palm oil. [2]

Percentage of unsaturated oil = %



- (iii) Oils such as palm oil can be refined and used to make soaps such as sodium oleate.

Complete the equation below, which represents the alkaline hydrolysis of the unsaturated oil. [2]



- (iv) Fats and oils are esters of carboxylic acids and glycerol.

Give the systematic name of glycerol (seen in part (iii) above). [1]

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END OF PAPER



[illegible]